

3D Coordinates · Lines & Rectangles on the Wall (x, y) — English Worksheet (Beginner)

Topic: Lines & Rectangles on the Wall (x, y) · **Course:** 3D Coordinates · **Level:** Beginner · **Time:** about 30 minutes

This lesson you use a **loop** to draw a whole **line** of blocks. Grow **x** to go across, or **y** to go up.

1 · Predict 🧐

Read the steps. Circle your answer.

```
set x to 0
repeat 4 times:
  place stone block at (x, 0)
  add 1 to x
```

x grows, y stays 0. A line across, or up? Circle one: across · up

Why? Circle one: x is the across number · y is the across number

```
set y to 0
repeat 4 times:
  place stone block at (0, y)
  add 1 to y
```

This time y grows, x stays 0. Across or up? Circle one: across · up

2 · Find the Difference 🐛

Each pair shows clean code, then a buggy version. Circle the bug.

Pair A — The line should go **across** the bottom of the wall.

```
# clean
set x to 0
repeat 4 times:
  place block at (x, 0)
  add 1 to x
```

```
# buggy
set x to 0
repeat 4 times:
  place block at (0, x)
  add 1 to x
```

Which way does the buggy line go? Circle one: up · across

Pair B — The line should be **4 blocks long**.

```
# clean
set x to 0
repeat 4 times:
  place block at (x, 0)
  add 1 to x
```

```
# buggy
set x to 0
repeat 2 times:
  place block at (x, 0)
  add 1 to x
```

How long is the buggy line? Circle one: 2 blocks · 4 blocks

3 · Fill the Gap

You want a line going **straight up** the left edge. x must stay **0** while y grows.

```
set y to 0
repeat 4 times:
  place block at (____, y)
  add 1 to y
```

Which number is missing? Circle one: 0 · y · 4

4 · Show Your Work 📷 🎥

Switch to your homework world. Use a loop to draw **one line going across** and another for **one line going up**, so they meet at the corner (0, 0) and make an **L** shape.

Record **one video** — one take, no stopping (a phone is fine). Show these in order:

1 · Your code. Scroll through it. Say what each part does.

2 · Your build. Point the camera. Name the parts.

Say these out loud, filling in the blanks:

Today I built _____.

I built it using this code: _____.

In this code I used _____.

The hardest part was _____.

That part was hard because _____.

The most fun part was _____.

Something new I learned was _____.

Submit ✅

Send this worksheet + your video to teacher on KakaoTalk.